

Chapter 4: Market Forces of Supply and Demand

Demand

The demand curve slopes downward, because other things equal, a lower price means a greater quantity demanded. The demand curve shows how price affects quantity demanded, holding constant everything else that affects buyers.

- A **change in price** moves us along a **fixed demand curve**
- A change in anything else that affects demand, income, the price of related goods, tastes, expectations, or the number of buyers, **shifts the entire curve** to a new position

Shifts in Demand

Any change that raises the quantity demanded at every price shifts the curve to the **right**, which is an **increase in demand**. Any change that lowers the quantity demanded at every price shifts the curve to the **left**, which is a **decrease in demand**.

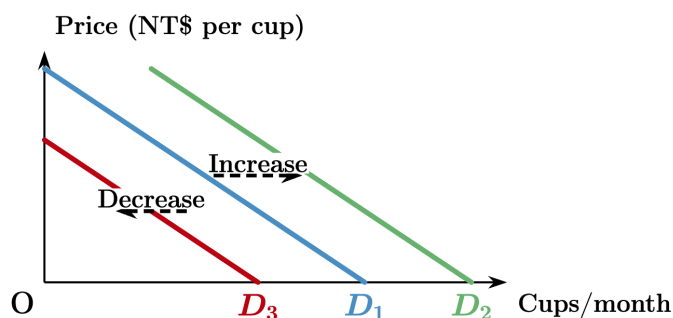


Figure 7: Shifts in the demand curve

What shifts the demand curve?

- **Income**: For a *normal good*, higher income raises demand (e.g. diamonds); for an *inferior good*, higher income lowers demand (e.g. instant ramen)
- **Prices of related goods**: A higher price of a **substitute** (e.g. coffee, for bubble tea) raises demand; a higher price for a **complement** (e.g. tapioca pearls, for bubble tea shops' costs) lowers demand for the other good
- **Tastes**: Anything that makes a good more or less appealing shifts demand in the same direction
- **Expectations**: Expecting **higher prices or higher income in the future** can raise demand today
- **Number of buyers**: More buyers in the market raises demand at every price

Normal and Inferior Goods

Most goods are **normal goods**. For example, if higher income leads a student to buy more bubble tea at each price, then bubble tea is a normal good for that student.

An **inferior good** is the *exception*. As a student's income rises, they may buy instant ramen less, because they may eat proper food more.

Case Study: Two Way to Reduce Smoking

As smoking causes illness, governments often want people to smoke less. This model provides two options:

1. **Shift the demand curve** for cigarettes to the left (e.g. graphic health warnings on packs, bans on tobacco advertising, and public health campaigns $\Rightarrow$ try to reduce the quantity demanded at any price)
2. **Raise the price** (move **along the demand curve**): E.g. Taiwan raised its tobacco tax by NT\$20 per pack, to the extent that it raises the price buyers pay, hence it reduces quantity demanded, while other demand factors are fixed

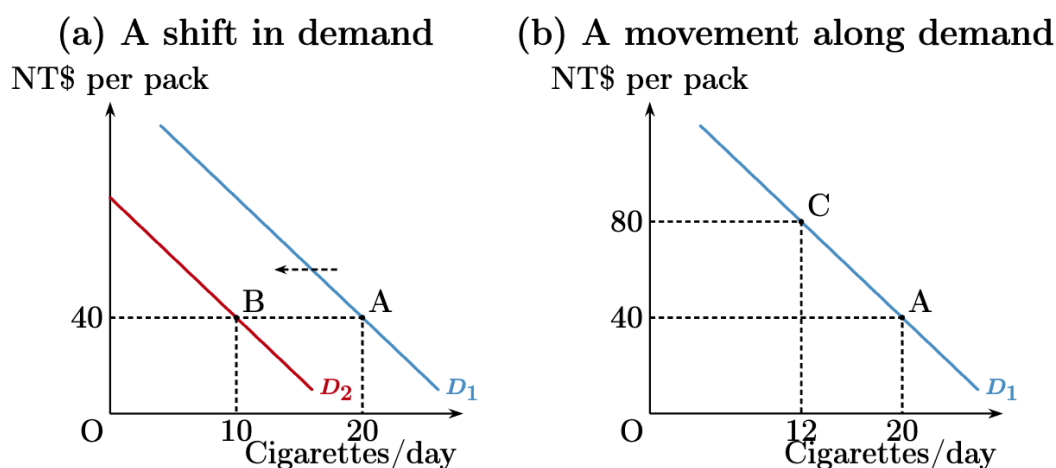


Figure 8: Diagrams representing the shift in the demand curve for the examples above

Supply

The supply curve slopes **upward** because, other things being equal, a higher price means a greater quantity supplied.

Shifts in Supply

Any change that raises the quantity supplied at every price shifts the curve to the **right**, an **increase in supply**. Any change that lowers the quantity supplied at any price shifts the curve to the **left**, a **decrease in supply**.

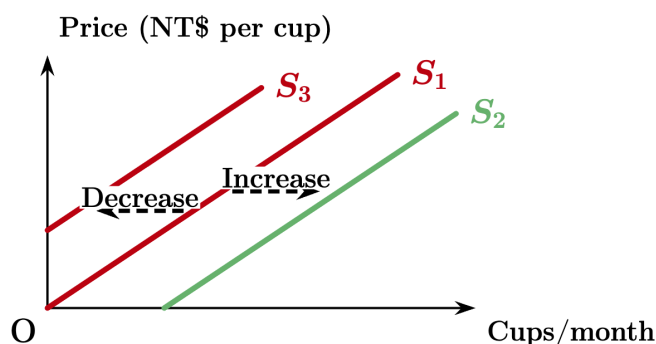


Figure 9: Shifts in the supply curve

What shifts the supply curve?

- **Input prices**: A higher price for an input makes production less profitable and shifts supply to the left
- **Technology**: An improvement that lowers production costs shifts supply to the right
- **Expectations**: Expecting a higher price tomorrow can lower supply today, as sellers of storable goods hold back inventory
- **Number of sellers**: More sellers in the market raises supply at every price, and sellers leaving the market lowers it

Equilibrium

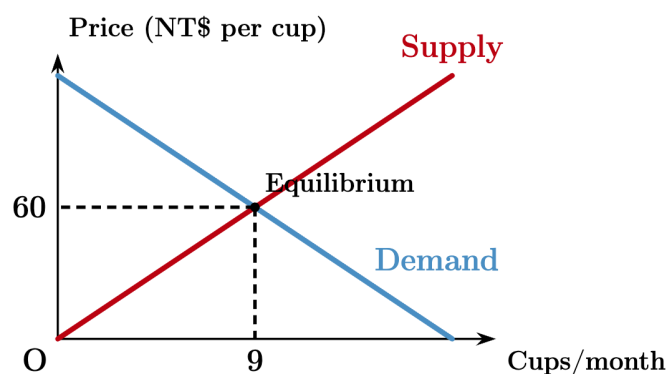


Figure 10: Equilibrium as a graph

Surplus

Key Idea

When there is a surplus of a good, sellers would **lower their price**, which raises quantity demanded and lowers quantity supplied, until the price reaches equilibrium.

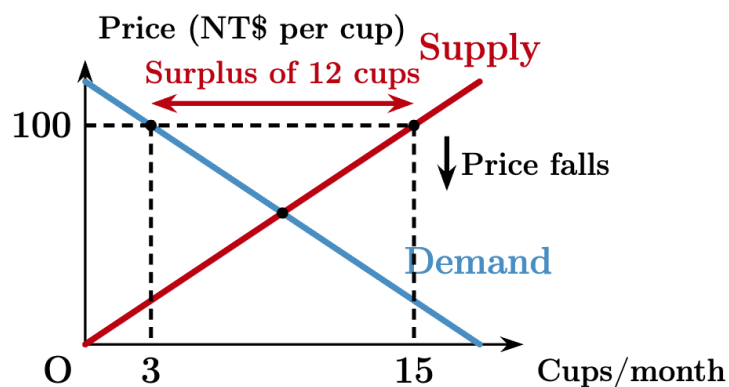


Figure 11: Surplus as a graph

Shortage

Key Idea

When there is a shortage of a good, sellers would **raise their price**, which lowers quantity demanded and increases quantity supplied, until the price reaches equilibrium.

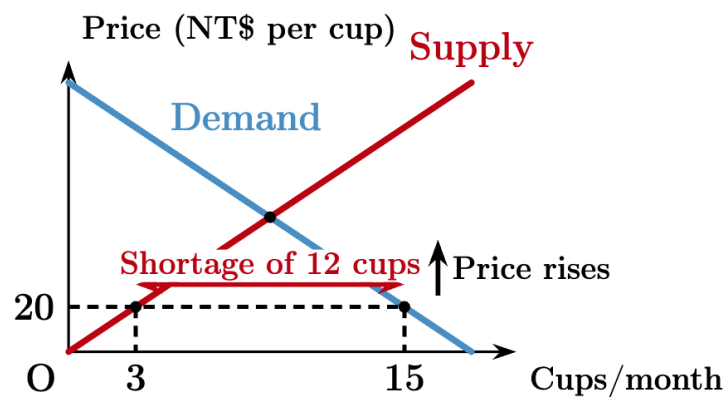


Figure 12: Shortage as a graph

Law of Supply and Demand

Key Idea

At equilibrium, there is no excess demand or supply pushing the price to change, but a new shock can shift the curve and change the equilibrium. **Adjustment takes time**. A binding price control can prevent the price from *clearing* the market, so a shortage or surplus may persist

How Prices Allocate Resources

Example of Price Gouging: Price Gouging and Taiwan's Mask Rationing

Even though the demand for masks increased due to the pandemic, the government didn't want the supply to simply increase and cause the prices to increase. This effect is known as price gouging.

Thus, Taiwan distributed masks, as requested by the government, through pharmacies using National Health Insurance cards. Taiwan also increased production to ease scarcity. This allowed for **higher supply** of masks, which makes the new equilibrium point of maintaining the same price to be possible.

Rationing allowed scarce masks without relying only on willingness to pay.

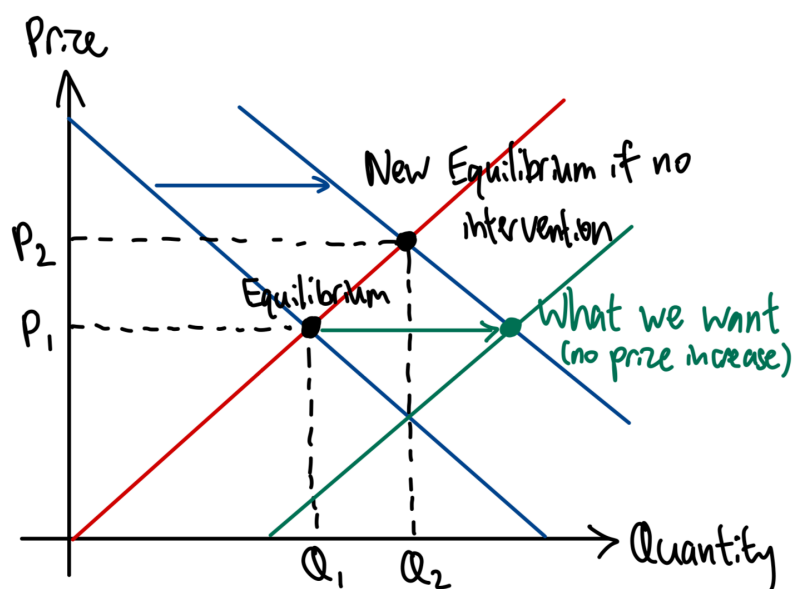


Figure 13: Mask gouging example's supply and demand

Key Idea

Market prices coordinate decisions about **how scarce resources** are used.

Three Steps for Analyzing a Change

When some *event affects a market*, the equilibrium changes, and we get a new price and a new quantity.

1. Decide whether the event shifts the supply curve, demanded curve, or both
2. Decide which direction each affected curve shifts
3. Compare the new equilibrium to the old one using the supply-and-demand diagram

Note that sometimes the shift can be *ambiguous*.